

Abhijit Deshpande

Applied AI/ML Engineer · RAG, Agents, LLM Fine-Tuning

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SUMMARY

Applied AI/ML Engineer who builds and ships production LLM systems: a LoRA fine-tuned intent classifier at 87% macro F1 with 94% lower inference cost, a LangGraph agent at 89% F1 under 800ms p95, and an end-to-end RAG pipeline (PGVector, cross-encoder reranking) tuned against a RAGAS evaluation harness. Production experience on an AWS Bedrock LLM root-cause system, on a foundation of classical ML, statistics, and design of experiments.

EXPERIENCE

Charter Communications | St. Louis, MO

Business Systems Analyst (AI/ML Systems Team) | Mar 2023 - Present

- Shipped a production AIOps platform targeting 60% lower MTTR, predicting database failures 15 to 60 minutes ahead and drafting the root-cause report before impact.
- Caught 71% of incidents across 12 database instances with a LightGBM early-warning model, engineering 14 telemetry signals into windowed binary classification with a lead-time gap to prevent label leakage.
- Defined the diagnosis path: MongoDB vector search retrieves comparable past incidents, then Amazon Bedrock Nova generates a grounded root-cause report with low-confidence re-analysis.
- Ran incident-free across 6 production servers for 9+ weeks, designing the evaluation framework (groundedness, evidence-traceability, false positives) and human-in-the-loop gates for a LangGraph multi-agent workflow.

University of Texas Arlington | Arlington, TX

Research Analyst - Machine Learning & AI | Jan 2021 - Mar 2023

- Built a Transformer-LSTM hybrid in PyTorch for intraday trading signals across 9 equities; achieved 0.64 AUC vs a 0.57 Random Forest baseline, stable across 16 window configurations.
- Designed a finance-specific labeling pipeline (Dollar-Volume Bars, Triple Barrier Method) with Purged K-Fold CV to prevent temporal leakage; tuned via Optuna Bayesian search (50+ trials) with PCA over 46 technical indicators.

PROJECTS

AURA: Agentic User Resolution Assistant | LangGraph · DeBERTa · AWS · Docker · Python [\[link\]](#)

- Built and deployed a hybrid conversational AI for customer support (Q&A, complaints, churn-retention) using LangGraph and AWS Lambda/API Gateway with DeBERTa intent routing; 89% 3-class F1 at <800ms p95.
- Replaced a fully autonomous agent that hallucinated tool arguments and looped indefinitely with hybrid deterministic routing plus scoped LLM tool selection.

INTELLA: Intent Classification with LoRA-Fine-Tuned Language Assistant | PyTorch · AWS SageMaker [\[link\]](#)

- Built a two-stage support-automation system for 150-intent chat classification: LoRA-fine-tuned RoBERTa, then 20-bucket intent grouping for FLAN-T5 generation (addressing sparse generation supervision).
- Achieved 87% macro F1 (+22pp over baseline) on 8K examples while cutting inference cost 94% vs GPT-3.5 (\$0.00012 vs \$0.002/query), projecting ~\$1.7K/month savings at 10K daily queries

IntelliQA: RAG-Powered Document Intelligence | Python · LangChain · PGVector · RAGAS · Docker · AWS [\[link\]](#)

- Built an end-to-end RAG backend: Tika parsing with format-aware chunking (heading, code, row-wise) and content-hash dedup, bge-large embeddings in PGVector (HNSW, ef_search 40 to 200), dense k=20 reranked by cross-encoder to top-8, and temperature=0 generation that refuses on insufficient context.
- Designed a 50-question RAGAS harness (faithfulness, relevancy, context precision/recall) over a mixed-domain corpus, tuning one variable at a time: recall 0.50 to 0.71 via k, precision 0.35 to 0.41 via reranking; built hybrid retrieval (RRF) and disabled it after evals showed no gain.

Credit Risk & Default Prediction | Python · XGBoost · SMOTE · scikit-learn · SQL [\[link\]](#)

- Built a default-risk pipeline on 150K loan records (6.7% default rate); 12 engineered features lifted AUC 0.08, and SMOTE + XGBoost raised minority-class recall 31% to 78%, catching 47pp more defaulters pre-default than baseline.

EDUCATION

University of Texas Arlington | **M.S. Industrial Engineering (GPA 3.9/4.0)** | May 2022

Focus: Design of Experiments, Statistical Methods, Regression Analysis, ML · IISE QCRE Data Challenge - Runner-Up

Shivaji University, India | **B.E. Mechanical Engineering** | May 2018

SKILLS

ML & AI:	PyTorch, HuggingFace, Scikit-learn, LoRA/PEFT, RAG, LangChain, LangGraph, RAGAS
Cloud & MLOps:	AWS (Bedrock, SageMaker, Lambda, API Gateway, S3, CloudWatch), Docker, Git, CI/CD
Data & Backend:	Python, SQL, Pandas, NumPy, Vector DBs, Django, Tableau
Statistics:	Hypothesis Testing, ANOVA, Bayesian Optimization, Design of Experiments, Time Series, SAS

CERTIFICATIONS

AWS Solutions Architect Associate, Deep Learning Specialization (DeepLearning.AI/Coursera), Tableau Desktop Specialist